VENKATA SUMANTH RAYASAM

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PROFESSIONAL SUMMARY

AI/ML Engineering graduate with hands-on experience building and training ML models for structured data use cases. Skilled in Python (Pandas, NumPy, Scikit-learn), data preprocessing, feature engineering on large datasets. Strong fundamentals in supervised/unsupervised learning algorithms, statistics, and model evaluation & tuning. Experienced translating business problems into ML solutions. Passionate about solving business problems using data-driven solutions.

TECHNICAL SKILLS

- **Programming Languages**: Python, SQL
- **Machine Learning:** Supervised & Unsupervised Learning Algorithms, Model Development & Evaluation
- **Libraries & Frameworks:** Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, FastAPI
- **Databases & Cloud Platforms:** MySQL, Amazon Web Services (AWS)
- **Core CS Fundamentals:** Object-Oriented Programming, Computer Networks, Data Structures & Algorithms
- **Tools & Version Control:** Git, GitHub, Visual Studio Code, Google Colab, Microsoft Office

PROJECTS

Non-Invasive Detection of Coronary Artery Disease ML Pipeline (Research Project), [GitHub Link]
Technologies Used: Python, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Claude AI

- **Built an end-to-end ML pipeline** for Coronary Artery Disease prediction on 1,200 + physiological records, including data preprocessing and feature engineering on structured healthcare data.
- **Applied feature engineering signal processing techniques** to extract meaningful patterns from physiological signals, improving model robustness and performance.
- **Built a stacked ensemble model** using XGBoost, Random Forest, LightGBM, and Logistic Regression to optimize predictive accuracy.
- **Evaluated and tuned models using standard performance metrics**, achieving 91% ROC-AUC, 88% accuracy.

Handwritten Digit Recognition System, [GitHub Link]
Technologies Used: Python, NumPy, Pandas, Tkinter, TensorFlow, Keras, Matplotlib

- **Developed a CNN-based deep learning model** using TensorFlow, achieving 98% accuracy on 60,000+ MNIST samples.
- Designed an interactive Tkinter GUI enabling real-time digit drawing and prediction with live confidence scores and a probability distribution bar chart across all 10 digit classes.

PUBLICATIONS

Reddy, R., Kauser, S.H., Rayasam, V.S. — *"Non-Invasive Coronary Artery Disease Screening from Photoplethysmography Using SHAP-Guided Feature Selection and Demographic Late Fusion"* — Computer Methods and Programs in Biomedicine (Elsevier), Under Review, 2026.

- Contributed to research methodology, model validation, and manuscript writing for a CAD screening pipeline achieving AUC 0.95 on 1,278 patients from the MIMIC-III clinical database.

EDUCATION

Bachelor of Technology in Artificial Intelligence & Machine Learning — 2023 – 2026
PACE Institute of Technology and Sciences, Valluru — **CGPA:** 8.8

CERTIFICATIONS

- **Python Programming**, Scalar Academy — June – 2025
- **AWS Academy Graduate – Cloud Architecting**, AWS, [Credential URL] — March – 2026
- **AWS Academy Graduate – Machine Learning Foundations**, AWS, [Credential URL] — March – 2024

ACHIEVEMENTS

- Achieved **Senior Under Officer (SUO)** rank in NCC — led and mentored 50+ cadets
- Participated in college level hackathon, Built an optimized ML solution under time pressure.